

Tianruo Rose Xu

Mobile: +1 607-262-1199 Email: tx88@cornell.edu Website: xtrose29.github.io

Education

Cornell University, College of Engineering

Aug 2024 – May 2027 (expected)

B.S. in Computer Science Minor in Mathematics

Selected coursework: Natural Language Processing, Deep Learning, Machine Learning, Reinforcement Learning, AI Practicum, Algorithms, Data Structures and Functional Programming, Computer Systems, Mathematical Foundations of Computing, Applied Logic, Linear Algebra, Probability and Statistics, Multivariable Calculus, ODEs, Number Theory, Mathematical Modeling.

Honors and Awards

- Hunter R. Rawlings III Cornell Presidential Research Scholar Award.
- Menschel Cornell Commitment Public Service Grant.
- Clare Boothe Luce Undergraduate Research Award.
- Bowers Undergraduate Research Experience (BURE) stipend and Academic Year Expansion support.
- USA Computing Olympiad (USACO) Platinum Division; perfect score in the [January 2023 Gold Contest](#).
- USA Mathematical Olympiad (USAMO) Qualifier.

Publications

[Submitted] *RT-Safe: Benchmarking Safety of Agent in Real-Time Embodied Environment*.

[Submitted] *Can Emergent Verbalization of Epistemic Uncertainty Inform Factuality?*

[Preprint] Haoqiang Kang, Yiming Zhang, Yiyang Guo, Chuying Li, Jianzhi Shen, **Tianruo Rose Xu**, Xiaokang Ye, Lianhui Qin. *DeliveryGym: An RL Environment for Long-Horizon Embodied Agent Planning with Adaptive Curriculum*. arXiv preprint arXiv:2609.19801, September 2026. [\[Paper\]](#) [\[Code\]](#).

[Published] **Tianruo Rose Xu**. *Can You Trust What I Think? Analyzing and Improving Verbalized Uncertainty and Factuality in Reasoning-Based Large Language Models*. AAAI-26 Undergraduate Consortium, Singapore, January 2026. [\[Poster\]](#) [\[Oral Presentation\]](#) [\[Paper\]](#).

[Published] **Tianruo Rose Xu**, Vedant Gaur, Liu Leqi, Tanya Goyal. *The Progress Illusion: Revisiting Meta-Evaluation Standards of LLM Evaluators*. Findings of the Association for Computational Linguistics: EMNLP 2025, Suzhou, China, November 2025. [\[Paper\]](#) [\[Poster\]](#).

[Published] Yingke Ding, Jiankai Tang, Wanying Mo, **Tianruo Rose Xu**, Yuanchun Shi, Yuntao Wang. *NFCMTL: Auto NailFold Capillaroscopy through a Multi-Task Learning Model*. MICCAI 2025 Workshop MSB EMERGE, July 21, 2025. [\[Paper\]](#) [\[Code\]](#). Runner-up Best Paper, MICCAI 2025 Workshop MSB EMERGE.

[Published] Tianruo Xu. *Elder Depression Detection by Multimodal Means*. BCP Education & Psychology, November 7, 2022. [\[Paper\]](#).

Research Experience

Undergraduate Researcher, Cornell University

Computer Science Department, NLP Group; advised by Professor Tanya Goyal and Claire Cardie.

Analyzing Verbalized Uncertainty and Factuality in Reasoning-Based LLMs

2025 – 2026

- Investigates whether verbalized epistemic uncertainty in LLM reasoning traces predicts fact-level factuality in long-form question answering.
- Led project design with Anirudh Atmakuru and Sanjana Ramprasad; built a pipeline for trace analysis, atomic fact extraction, trace-response alignment, FactScore evaluation, large-scale empirical analysis, and classifier training.

Revisiting Meta-Evaluation Standards of LLM Evaluators

2025

- Develops fine-grained meta-evaluation methods for auditing LLM-as-judge benchmarks, evaluator reliability, and reference-model sensitivity.
- Led the evaluation framework design; implemented data collection and scoring pipelines; ran large-scale experiments; analyzed correlations with human judgments; contributed to writing, figures, and result interpretation.

Content and Style Evaluation in Natural Language Generation Systems 2024

- Studies content and style as separable evaluation dimensions in natural language generation systems.
- Co-supervised by Professor Claire Cardie; designed experiments, implemented data processing and evaluation pipelines, ran model-based evaluations, conducted error analysis and ablations, and contributed to writing and result visualization.

Undergraduate Researcher, Cornell University

Mathematics Department, Math + AI Lab; advised by Professor Daniel Halpern-Leistner.

Lean Proof Arena and LaTeX-to-Lean Autoformalization Pipeline 2026 – present

- Develops a DARPA ExPMath-supported platform and pipeline for research-level autoformalization, centered on Lean Proof Arena and LaTeX-to-Lean conversion.
- Building Lean Proof Arena for comparing formal proof attempts; developing a LaTeX-to-Lean pipeline, benchmark tasks, and evaluation protocols for formal correctness and semantic fidelity.

Research Intern, UC San Diego

Computer Science Department, Q-Lab; advised by Professor Lianhui Qin.

RT-Safe: Real-Time Embodied Agent Safety 2026 – present

- Lead RT-Safe, a benchmark evaluating embodied agent safety in dynamic environments that continue evolving during model inference; manuscript submitted.
- Defined the research direction, safety scenario taxonomy, benchmark tasks, and evaluation workflow; led analysis of how inference latency and reasoning effort affect navigation safety.

DeliveryGym / DeliveryBench: Embodied Agent Environments 2026

- Developed visual traffic, obstacle, and safety infrastructure for VAGEN/DeliveryBench, including pedestrian traffic-light rendering, signal-state checks, rollout support, and hazard evaluation tools.
- Built and maintained the Unreal Engine-rendered first-person-view observation pipeline and assets, including traffic-signal and blocked-road variants.
- Extended DeliveryBench with the CityCore Paris map, integrating semantic points of interest, buildings, road and navigation graphs, coordinate transforms, and 2D map assets; contributed CityCore Paris visual materials.

Undergraduate Research Assistant, Tsinghua University

Department of Computer Science and Technology, Pervasive Human-Computer Interaction Lab.

Intelligent AI Tool Benchmarking and Orchestration Platform 2025 – present

- Develops an intelligent AI tool platform that standardizes tool capability descriptions, matches tools to user intents, and supports multi-tool workflow orchestration.
- Designed and implemented back-end search and matching algorithms for tool retrieval, user-intent matching, and personalized AI tool selection.

NFCMTL: Auto NailFold Capillaroscopy through a Multi-Task Learning Model 2025

- Develops a multi-task learning model for automated nailfold capillaroscopy, combining segmentation, classification, and keypoint detection for non-invasive microvascular image analysis.
- Conducted dataset preprocessing, baseline training, experiment execution, model evaluation, and preliminary result analysis to support development of the NFCMTL framework.

Undergraduate Research Assistant, Cornell University 2024

Information Science Department; advised by Keigo Kusumegi and Professor Yian Yin.

- Studied how diverse collaboration patterns affect Wikipedia content quality.
- Processed Wikipedia data and applied K-means clustering to analyze article-quality patterns.

Teaching Experience

Cornell University

- **Head Teaching Assistant**, CS 3780: Introduction to Machine Learning Fall 2026
- **Climbing Instructor**, PE 1640: Basic Rock Climbing Fall 2026
- **Teaching Assistant**, CS 3780: Introduction to Machine Learning Spring 2026
Designed the final Kaggle challenge, [All About Birds!](#), a bird-sound spectrogram classification task.

- **Teaching Assistant**, CS 4820: Introduction to Analysis of Algorithms
 - **Teaching Assistant**, CS 2800: Mathematical Foundations of Computing
- Fall 2025
Spring 2025

Industry Experience

Incoming Machine Learning Engineer Intern, TikTok

TikTok E-commerce Knowledge Graph; NLP/CV/Multimodal/Algorithm.

2027
Seattle, WA

Independent Research Collaboration, Oracle

Remote; mentored by Priyan Pattnayak, Senior Principal Scientist at Oracle Cloud.

Aug 2026 – present